SAGAR MANNA

Software Engineer

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EXPERIENCE

AI Engineer

Vedas Tech

June 2026 – Present | **Gujarat, India**

- Engineered modular **AI/ML inference pipelines** for voice cloning and multilingual TTS, improving model integration and reusability.
- Integrated **NLP-based text processing** with speech synthesis workflows to support multilingual text-to-audio generation.
- Built asynchronous AI processing using **Python, FastAPI, Redis, and Celery**, enabling non-blocking inference workflows.
- Developed reusable audio preprocessing and model-serving components, reducing code duplication and maintenance overhead.
- Designed intelligent geospatial search and location-matching algorithms to improve search relevance and discovery efficiency.
- Optimized **REST APIs** and database operations for efficient data retrieval and backend performance.
- Designed modular AI service architecture to simplify model integration, testing, and deployment readiness.
- **Tech Stack:** Python, NLP, AI/ML, Model Inference, Multilingual TTS, FastAPI, Redis, Celery, PostgreSQL, SQLite, REST APIs, Git, GitHub.

Frontend Developer Intern

PearlThoughts

FEB 2026 - April 2026 | **Remote, India**

- Refactored a monolithic Email Card component into modular **React** components, improving code maintainability and reuse.
- Migrated legacy class-based UI to **functional components** using **React Hooks**, modernizing component architecture.
- Reduced code duplication by 30% by extracting shared, reusable UI components across the application.
- Built responsive, cross-device UI layouts using **TailwindCSS** and utility-first styling principles.
- Managed component-level state and data flow using **React Hooks** and prop drilling patterns.
- **Tech Stack:** React.js, Next.js, JavaScript (ES6+), TailwindCSS, Git, GitHub

TECHNICAL SKILLS

- **Languages:** JavaScript (ES6+), TypeScript, Python, C++, C
- **Frontend:** React.js, Next.js, HTML5, CSS3, Tailwind CSS, React Hooks, Context API, Responsive Web Design
- **Backend:** Node.js, Express.js, FastAPI, RESTful APIs, JWT Authentication, Nodemailer, Multer
- **Databases:** PostgreSQL, MongoDB, MySQL
- **AI Agentic Systems:** Agentic AI, Multi-Agent Systems, LangChain, LangGraph, LLMOps, OpenAI API
- **DevOps Developer Tools:** Git, GitHub, CI/CD, Postman
- **Certifications:** DotNet Full Stack — Wipro TalentNext

EDUCATION

B.Tech. (CSE) - 9.0 SGPA

Chandigarh Engineering College,Landran

2023 – 2026 | Mohali, Punjab

PROJECTS

BharashaRide – Driver Alert System

- **Tech Stack:** ReactJS | FastAPI | OpenCV | Dlib | Python | JavaScript
- Built a real-time driver monitoring system detecting drowsiness, yawning, and emotions using facial landmark tracking.
- Triggered instant audio alerts and safety warnings based on ML-driven thresholds.
- Integrated React dashboard with FastAPI backend for live alerts and visualization.
- Github

ResearchMind – Multi-Agent Research

- **Tech Stack:** Python | Streamlit | LangChain | LangGraph | OpenAI API
- Built an AI-powered multi-agent system for automated research and report generation.
- Integrated Tavily and BeautifulSoup for real-time web search and data extraction.
- Developed LangChain writer and critic chains for report refinement and scoring.
- **Impact:** Reduced manual research effort by 40% through AI automation.
- Github

MoneyTracker – Expense Management System

- **Tech Stack:** React.js | Node.js | Express.js | MongoDB
- Built a full-stack expense tracking application using the MERN stack.
- Developed secure REST APIs with real-time balance and transaction management.
- Integrated MongoDB Atlas for scalable cloud-based data storage.
- **Impact:** Improved expense tracking efficiency and user experience by 30%.
- Github

Room Booking Dashboard

- **Tech Stack:** Next.js | React | Tailwind CSS | TypeScript
- Built a responsive hotel room booking platform with search and filtering features.
- Added booking flow, payment simulation, booking history, and admin dashboard.
- Developed reusable UI components with clean state management and API handling.
- Improved booking experience and UI responsiveness by 35%.
- Github